SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Duration: 60 Mins **QUESTIONS: 5** **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I MONISH.L [1972260035] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Marks/ 10
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty *Total Marks Awarded:*

Annexure A

Scenario-Based

1. A newly established engineering college is assigned the IP block **192.168.100.0/24** to design its internal network. The network administrator has already assigned subnet masks for each department as follows: Computer Science → /25, Electronics → /26, and Mechanical → /27. Using the given subnet masks, apply subnetting techniques to determine the network address and broadcast address for each department. Identify the valid range of host IP addresses for each subnet. Calculate the number of usable hosts available in each subnet. Finally, determine the remaining unused IP address range within the given network block. **(Marks 10)**

2. A software company is provided with the IP block **10.10.0.0/16** for its internal network. The network engineer has already assigned subnet masks for each team as follows: Development → /24, Testing → /25, HR → /26, and Management → /27. Using the given subnet masks, apply subnetting techniques to determine the network address and broadcast address for each team. Identify the valid range of host IP addresses for each subnet. Calculate the number of usable hosts available in each subnet. Finally, determine the remaining unused IP address range within the given IP block. **(Marks 10)**

3.An organization is assigned the IPv6 address **2001:0db8:abcd:0000:0000:0000:0000:0000/64** for its network configuration. The network administrator requires both compressed and expanded forms of the given IPv6 address. Apply IPv6 addressing rules to convert the given address into

its compressed form and then expand it back to its full notation. Using the prefix length **/64**, determine the network prefix of the given address. Calculate the total number of possible host addresses available within this network. **(Marks 10)**

4. A company has interconnected three branch offices using routers with the subnets **192.168.1.0/24**, **192.168.2.0/24**, and **192.168.3.0/24**. A data packet arrives at the central router with the destination IP address **192.168.2.45**. Using the given subnet information, apply routing techniques to determine the appropriate network to which the destination IP belongs. Identify the matching route entry based on the destination address. Determine the next-hop network for forwarding the packet. Also, identify the default route that can be used for destinations not present in the routing table. **(Marks 10)**

5. A small office network consists of two LANs connected through a router. The first LAN uses the subnet **192.168.10.0/26**, and the second LAN uses **192.168.10.64/26**. The network administrator assigns IP addresses to the router interfaces as **192.168.10.1** for the first LAN and **192.168.10.65** for the second LAN. Using the given configuration, apply subnetting techniques to determine the valid host IP range for each LAN. Identify suitable IP addresses that can be assigned to computers in both LANs. Determine the network and broadcast address for each subnet. Also, identify the gateway address used by hosts in each LAN for communication. **(Marks 10)**

ANSWERS

Answer 1:

Given block: 192.168.100.0/24 (256 addresses, .0 to .255)
Allocating blocks in order (VLSM) – CS /25, Electronics /26, Mechanical /27:

Computer Science → /25 (126 usable hosts)
Network address : 192.168.100.0
Broadcast address : 192.168.100.127

Valid host range : 192.168.100.1 – 192.168.100.126
Usable hosts : 2^7 − 2 = 126

Electronics → /26 (62 usable hosts)
Network address : 192.168.100.128
Broadcast address : 192.168.100.191
Valid host range : 192.168.100.129 – 192.168.100.190
Usable hosts : 2^6 − 2 = 62

Mechanical → /27 (30 usable hosts)
Network address : 192.168.100.192
Broadcast address : 192.168.100.223
Valid host range : 192.168.100.193 – 192.168.100.222
Usable hosts : 2^5 − 2 = 30

Remaining unused block : 192.168.100.224 – 192.168.100.255 (192.168.100.224/27, 32 addresses free for future use)

Answer 2:

Given block: 10.10.0.0/16 (65536 addresses)

Allocating blocks in order (VLSM) – Development /24, Testing /25, HR /26, Management /27:

Development → /24 (254 usable hosts)
Network : 10.10.0.0 Broadcast : 10.10.0.255
Host range : 10.10.0.1 – 10.10.0.254
Usable hosts : 2^8 − 2 = 254

Testing → /25 (126 usable hosts)
Network : 10.10.1.0 Broadcast : 10.10.1.127
Host range : 10.10.1.1 – 10.10.1.126
Usable hosts : 2^7 − 2 = 126

HR → /26 (62 usable hosts)
Network : 10.10.1.128 Broadcast : 10.10.1.191
Host range : 10.10.1.129 – 10.10.1.190
Usable hosts : 2^6 − 2 = 62

Management → /27 (30 usable hosts)
Network : 10.10.1.192 Broadcast : 10.10.1.223
Host range : 10.10.1.193 – 10.10.1.222
Usable hosts : 2^5 − 2 = 30

Remaining unused range : 10.10.1.224 – 10.10.255.255 (all remaining space within the /16 block)

Answer 3:

Given address : 2001:0db8:abcd:0000:0000:0000:0000:0000 /64

Compressed form : 2001:db8:abcd::
(leading zeros in each group dropped, and the longest run of consecutive all-zero groups replaced with '::')

Expanded (full) form : 2001:0db8:abcd:0000:0000:0000:0000:0000

Network prefix (/64) : 2001:0db8:abcd:0000::/64 ≡ 2001:db8:abcd::/64

Total possible host addresses : 2^(128−64) = 2^64
= 18,446,744,073,709,551,616 host addresses

Answer 4:

Given subnets : 192.168.1.0/24 , 192.168.2.0/24 , 192.168.3.0/24

Destination IP : 192.168.2.45

Matching network : 192.168.2.0/24 — since .45 lies between .1 and .254 of this subnet

Matching route entry : 192.168.2.0/24 (longest-prefix match in the routing table)

Next-hop network : the interface/network directly attached to 192.168.2.0/24 — i.e. the router forwards the packet out on the LAN 2 interface (192.168.2.0/24) for final delivery

Default route : 0.0.0.0/0 — used to forward any destination address that does not match an entry in the routing table (e.g. traffic outside these three subnets)

Answer 5:

LAN 1 – 192.168.10.0/26

Network address : 192.168.10.0 Broadcast address : 192.168.10.63

Valid host range : 192.168.10.1 – 192.168.10.62

Gateway (router interface) : 192.168.10.1

Assignable to computers : 192.168.10.2 – 192.168.10.62 (192.168.10.1 reserved for the gateway)

LAN 2 – 192.168.10.64/26

Network address : 192.168.10.64 Broadcast address : 192.168.10.127

Valid host range : 192.168.10.65 – 192.168.10.126

Gateway (router interface) : 192.168.10.65

Assignable to computers : 192.168.10.66 – 192.168.10.126 (192.168.10.65 reserved for the gateway)